

MYCOttage

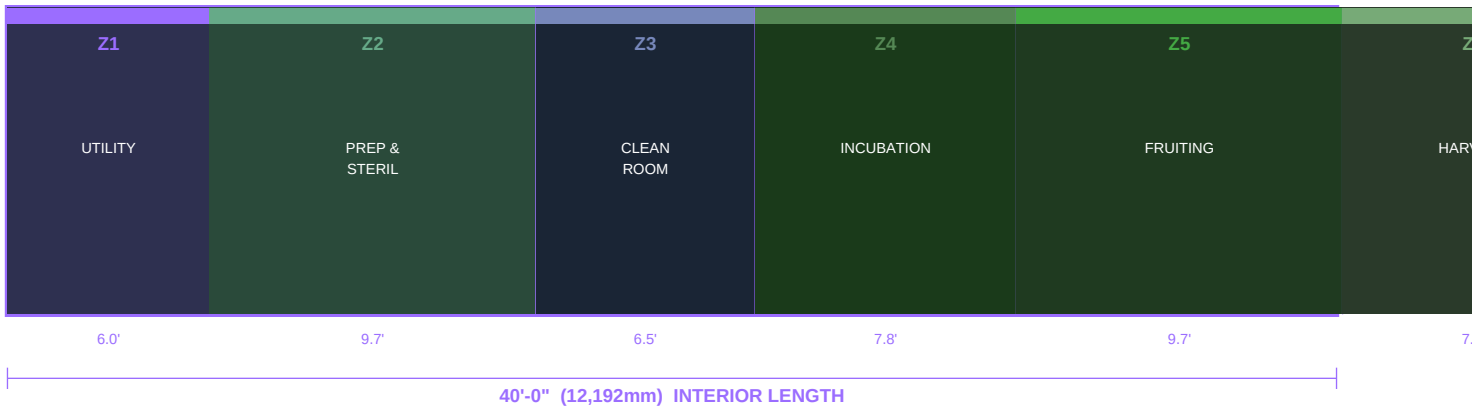
CONTAINERIZED MYCOLOGY PRODUCTION SYSTEM

CONTAINER	PRODUCTION	OUTPUT TARGET	INTEGRATION	INSTALL TYPE
40FT HIGH CUBE 9'6" INTERIOR HT	ALL STAGES FULL CYCLE	50-200 LBS/WK	GRUSH BATCH TRACKING	SITE BUILT ZONE 9B

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01 — DIMENSIONED FLOOR PLAN

40ft High Cube container interior: 39'5" × 7'8" × 8'10" usable (11,989mm × 2,337mm × 2,692mm). All dimensions to finished wall face after insulation and panel. Zones separated by removable partitions unless noted fixed.



Z#	ZONE NAME	LENGTH (FT/IN)	LENGTH (MM)	FUNCTION
Z1	Utility / Mechanical	6'-0"	1,829mm	AC, panel, water, CO2
Z2	Prep & Sterilization	9'-8"	2,946mm	Substrate, autoclave, sink
Z3	Clean Room / Inoculation	6'-6"	1,981mm	LAF hood, SAB, inoculation
Z4	Incubation	7'-9"	2,362mm	Dark racks, 75-80F, high CO2
Z5	Fruiting Chamber	9'-8"	2,946mm	FAE, misting, 3-rack array
Z6	Harvest & Processing	7'-9"	2,362mm	Trim, scale, cold storage

NOTES: Aisle centerline runs full length at 3'-10" from front wall. All zones have 7'-0" (2,134mm) clear height after HVAC ducting. Clean room (Z3) positive-pressure sealed with acrylic panel walls. Container orientation: utility end (Z1) away from prevailing wind.

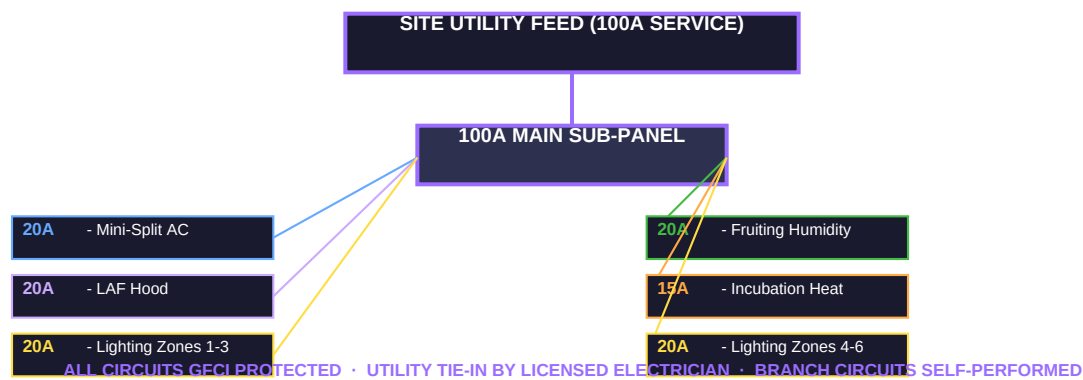
02 — ZONE SPECIFICATIONS

Z#	ZONE NAME	DESCRIPTION & KEY EQUIPMENT	PARTITION
Z1	Utility / Mechanical	18,000 BTU mini-split AC (wall mount). 100A sub-panel. 5-stage RO water system.	Fixed wall (plywood + FRP). CO2 tank optional.
Z2	Prep & Sterilization	Stainless work counter with sink (8ft). All American 25qt pressure cookers x2.	Open to aisle. Substrate mixing station. Bag cooling rack.
Z3	Clean Room / Inoculation	Laminar flow hood (2ft HEPA). Still air box (SAB) backup. UV-C sterilization lamp.	Acrylic panel + door seal. (interlocked off when occupied).
Z4	Incubation	NSF wire shelving x2 (6 shelves each, 74" posts). Wire mesh for airflow.	Blackout curtain. Blacked out, zero light penetration.
Z5	Fruiting Chamber	NSF wire shelving x3 (6 shelves each). Ultrasonic humidifiers x3 on Inkbird RH controller.	Removable panel. AC Infinity inline fans x2 for airflow.
Z6	Harvest & Processing	Harvest/trim table (stainless, 6ft). Digital scale (0.1g/30kg). Vacuum sealer.	Open to aisle. Impulse bag sealer. Upright refrigerator (30L).

GRUSH INTEGRATION: Scan points at Z3 (batch open/inoculation), Z4 entry (incubation load — shelf location logged), Z5 entry (fruiting transfer — stage change), and Z6 (harvest weight log — batch close or flush extension). Environmental sensor data feeds into batch records via Home Assistant hub or direct API.

03 — ELECTRICAL SINGLE-LINE DIAGRAM

100A service minimum. All branch circuits self-performed by licensed wireman. Utility tie-in to site meter by licensed electrician with permit. Container shell bonded to ground rod. All circuits GFCI protected (wet environment).



CIRCUIT	LOAD	BREAKER	NOTES
Mini-Split AC	18,000 BTU	20A 240V	Dedicated circuit. Drain to exterior.
Fruiting Humidity	3x ultrasonic hmdr	20A 120V	Inkbird controller on shared circuit.
LAF Hood	1.5A fan + HEPA	20A 120V	UV-C lamp interlocked off when door open.
Incubation Heat	Heat mat array	15A 120V	Inkbird temp controller.
Lighting Z1-Z3	LED strips	20A 120V	Timer controlled.
Lighting Z4-Z6	LED strips (660nm red)	20A 120V	Fruiting spectrum. Separate timer.
Outlets (general)	General use	20A 120V	GFCI. Min 2 per zone.

GROUNDING: Container shell must be bonded to a dedicated ground rod at install site. All metal conduit, panel enclosure, and equipment frames bonded to panel ground bus.

04 — EQUIPMENT & PARTS LIST

Mid-range pricing sourced from public retail listings as of June 2026. Prices vary by supplier and region. Verify current pricing before budgeting.

ZONE	ITEM	SPEC / MODEL	QTY	EST. COST
Z1	Mini-Split AC	Pioneer 18,000 BTU 230V	1	\$650-\$850
Z1	Sub-Panel	Square D 100A 20-space	1	\$120-\$180
Z1	RO Water System	5-stage, 75 GPD	1	\$150-\$220
Z1	CO2 Tank (optional)	20lb aluminum + regulator	1	\$120-\$180
Z2	Work Counter + Sink	Stainless, 8ft, NSF	1	\$350-\$500
Z2	Pressure Cooker	All American 25qt	2	\$220-\$280 ea
Z2	Bag Cooling Rack	Wire, freestanding	2	\$40-\$60 ea
Z3	LAF Hood	2ft HEPA, pre-built	1	\$350-\$500
Z3	Still Air Box	Sterilite 116qt tote mod	1	\$30-\$50
Z3	UV-C Lamp	Interlocked, 254nm	1	\$60-\$100
Z4	NSF Wire Shelving	74" posts, 6-shelf, 48"W	2	\$120-\$160 ea
Z5	NSF Wire Shelving	74" posts, 6-shelf, 48"W	3	\$120-\$160 ea
Z5	Ultrasonic Humidifier	6L, Inkbird RH controller	3	\$50-\$80 ea
Z5	Inline Fan (FAE)	AC Infinity CLOUDLINE S6	2	\$80-\$100 ea
Z6	Harvest Table	Stainless, 6ft	1	\$200-\$300
Z6	Digital Scale	0.1g / 30kg capacity	1	\$40-\$70
Z6	Vacuum Sealer	Chamber type preferred	1	\$150-\$400
Z6	Upright Refrigerator	34-38F, 10-14 cu ft	1	\$400-\$600
ALL	Epoxy Floor Coating	2-part, commercial grade	1	\$200-\$350
ALL	Spray Foam Insulation	2-3in closed cell, all walls	1	\$600-\$1,000
ALL	FRP Wall Panels	4x8 sheets, food-safe	20	\$35-\$55 ea
ALL	LED Strip Lighting	Waterproof, 5000K + 660nm	6	\$30-\$60 ea
ALL	Grush Scan Points	QR plates, anodized aluminum	4	\$8-\$12 ea

ESTIMATED BUILD TOTAL (equipment only, excluding container, site prep, and electrical labor): \$4,800 - \$7,200 depending on tier. See Cottage Series - Pricing Tiers v1.0 for full cost breakdown.

05 — ENVIRONMENTAL TARGETS

Target ranges for each zone. Sensors log to Home Assistant and export to Grush batch records. All targets assume Zone 9B ambient (summer highs 100-110F exterior).

ZONE	TEMP TARGET	RH %	CO2 (PPM)	FAE / AIRFLOW	LIGHTING
Z1 Utility	Ambient	--	--	Exhaust fan	LED utility
Z2 Prep	Ambient	40-50%	--	Exhaust fan	5000K LED
Z3 Clean Room	68-72F	40-50%	Ambient	Positive pressure	5000K + UV-C
Z4 Incubation	75-80F	60-70%	2,000-5,000	Low / passive	Blackout
Z5 Fruiting	65-72F	85-95%	500-1,000	Active timed FAE	12hr 660nm
Z6 Harvest	60-65F	50-60%	Ambient	Exhaust only	5000K LED
Z6 Cold Storage	34-38F	--	--	Fridge internal	Fridge internal

SENSOR INTEGRATION

SENSOR / DEVICE	LOCATION	DATA OUTPUT	INTEGRATION
SCD40 CO2/Temp/RH	Z4, Z5	I2C - Home Assistant	Grush batch record API
SHT31 Temp/RH	Z3, Z6	I2C - Home Assistant	Grush batch record API
Inkbird IBS-TH2	Z4 backup	Bluetooth	Manual log or HA bridge
Inkbird RH Controller	Z5	Relay output	Humidifier on/off control
AC Infinity Controller	Z5 FAE	Fan speed PWM	Timed schedule

06 — BUILD PHASE CHECKLIST

Sequential build order. Complete each phase before beginning the next. DIY-friendly tasks marked (DIY). Licensed trade required marked (LIC).

PHASE 1 — CONTAINER ACQUISITION & PREP

- Source 40HC from shipyard or dealer. Inspect: no fumigation residue, no major rust-through, doors functional. DIY
- Pressure wash interior. Treat all rust with converter. Allow to cure 48hr. DIY
- Apply 2-3in closed-cell spray foam to all walls, ceiling, and door interior. DIY/PRO
- Install vapor barrier over foam where foam is not self-skinning. DIY
- Apply epoxy floor coating (2-part). Allow 72hr cure before loading. DIY
- Install FRP wall panels over foam with construction adhesive and screws. DIY

PHASE 2 — UTILITY & ELECTRICAL ROUGH-IN

- Coordinate with licensed electrician for site utility feed to container pad. LIC
- Install 100A sub-panel in Z1. Bond panel enclosure to ground bus. LIC
- Run conduit and pull wire for all branch circuits per diagram. LIC/DIY
- Install GFCI breakers or GFCI outlets on all circuits. Test each. LIC
- Install equipment outlets in each zone. Label all circuits at panel. LIC/DIY
- Bond container shell to ground rod driven at site. LIC

PHASE 3 — HVAC & ENVIRONMENTAL SYSTEMS

- Mount mini-split indoor head in Z1. Run lineset to outdoor unit. PRO
- Commission mini-split. Verify cooling in fruiting and clean room zones. PRO
- Install AC Infinity inline fans for FAE in Z5. Run to exterior vent. DIY
- Install Inkbird RH controller and ultrasonic humidifiers in Z5. DIY
- Install positive pressure filtered supply to Z3 (LAF hood as primary). DIY
- Mount CO2 tank and regulator in Z1. Run line to Z4 if using enrichment. DIY
- Install all environmental sensors per zone. Wire to HA hub in Z1. DIY

PHASE 4 — PLUMBING & WATER

- Hire plumber — run site water supply line to container inlet. LIC
- Install RO system in Z1. Connect to site supply line. DIY/LIC
- Run RO output line to sink in Z2 and float valve in humidifier reservoir. DIY
- Install drain from sink. Run to site drain or dry well. LIC
- Test all water connections under pressure. Check for leaks. DIY

PHASE 5 — ZONE BUILDOUT & SHELVING

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| ■ Install NSF wire shelving in Z4 (x2 units) and Z5 (x3 units). | DIY |
| ■ Apply Grush QR scan labels to each shelf position. Log location IDs. | DIY |
| ■ Build or install acrylic panel walls for Z3 clean room. Install door seal. | DIY |
| ■ Install interlocked UV-C lamp in Z3. Wire interlock to door switch. | DIY |
| ■ Install LAF hood in Z3. Verify HEPA filter seated and fan operating. | DIY |
| ■ Install LED lighting in each zone per spec. Wire to timers. | DIY |
| ■ Install harvest table, scale, and sealing equipment in Z6. | DIY |
| ■ Install upright refrigerator in Z6. Allow 24hr to reach temp before use. | DIY |

PHASE 6 — GRUSH INTEGRATION & COMMISSIONING

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| ■ Configure Home Assistant hub in Z1. Connect all sensors. | DIY |
| ■ Set up Grush batch record API connection or CSV export pipeline. | DIY |
| ■ Run substrate batch through full workflow: Z2 > Z3 > Z4 > Z5 > Z6. | DIY |
| ■ Verify all scan points trigger correct batch record events. | DIY |
| ■ Log environmental readings for 72hr across all zones. Adjust targets. | DIY |
| ■ Document final zone temps, RH, and FAE settings in Grush build record. | DIY |
| ■ Photograph completed build. Update MYCOttage build record in Drive. | DIY |